

Jay Thakur

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SUMMARY

Integration Engineer with experience delivering healthcare interoperability and interface implementation projects across multi-system environments. Skilled in designing and deploying HL7 v2.5 interfaces (ADT, ORU, ORM, DFT, SIU), supporting end-to-end implementations including requirements gathering, system configuration, validation, go-live, and production support. Hands-on exposure to HL7 FHIR (R4) through self-driven work, with practical understanding of resource structures and API-based data exchange within US healthcare workflows. Strong client-facing experience with a focus on interoperability, data quality, and reliable system integration in fast-paced healthcare environments.

TECHNICAL SKILLS

Interoperability & Standards: HL7 v2.5 (ADT, ORU, ORM, DFT, SIU), HL7 FHIR (R4 – working knowledge), JSON, XML, CSV, ICD-10, SNOMED CT, ASTM, EDI

Integration & Connectivity: Mirth Connect, Allscripts Interface Engine, API integrations, SFTP, VPN, TCP/IP (MLLP), file-based integrations

Healthcare Systems: EMR/EHR (Allscripts), LIS, PACS/RIS, Practice Management (PM), healthcare workflow mapping, interoperability QA

Data & Analysis: SQL (Oracle, MySQL), Python, data validation, data quality assurance, ETL concepts

Delivery & Implementation: Requirements gathering, interface design & mapping, UAT support, production support, stakeholder management, escalation handling, change management

Compliance: HIPAA, GDPR

EXPERIENCE

Sutherland Healthcare Solutions

Implementation Engineer

Hyderabad, India

February 2022 – December 2022

- Lead technical implementation for Allscripts EMR/PM systems across 25+ US healthcare facilities, generating \$300k+ in project revenue.
- Designed and deployed HL7 v2.5 interfaces (ADT, ORM, ORU, DFT, SIU) to enable real-time clinical and administrative data exchange between EMR, LIS, and PACS/RIS systems.
- Configured secure data pipelines using MLLP, SFTP, and VPN protocols, managing both legacy file-based integrations and modern API-driven workflows (JSON/XML).
- Managed the full interface lifecycle: analyzed vendor specifications to create technical mapping documents, performed SQL-based data validation, and led UAT/Go-Live for complex multi-vendor environments.
- Utilized SQL for deep-dive root cause analysis and message troubleshooting, reducing recurring production integration issues by 30%.
- Ensured 100% adherence to HIPAA and GDPR standards for patient privacy and secure data exchange during large-scale migrations.

Sigma AI Data Ltd

Data Annotation

Hammersmith, UK

October 2024 – April 2026

- Performed large-scale data validation and classification to support AI model training, consistently maintaining error rates below 5%.
- Processed high-volume datasets (2,500+ daily), strengthening quality workflows and issue identification practices aligned with technical service environments.

Kantar Analytics Practice

Data Analyst

Bangalore, India

April 2021 – July 2021

- Analyzed large campaign, customer, and sales datasets to identify trends, patterns, and growth opportunities, improving performance across multiple client engagements.
- Applied statistical techniques such as regression analysis, hypothesis testing, forecasting, and segmentation to enhance

reporting accuracy and support strategic decision-making.

- Built and leveraged Marketing Mix Models (MMM) to evaluate channel performance, optimize budget allocation, and measure media impact on revenue and ROI.
- Developed dashboards, KPI reports, and automated reporting processes to track campaign effectiveness, customer behavior, and key business outcomes.
- Delivered actionable insights and data-driven recommendations in collaboration with cross-functional teams, driving revenue growth, improved retention, and stronger marketing ROI.

PROJECTS

Healthcare Integration Project – Allscripts EMR Implementation

HL7, EMR, PACS/RIS, SQL

Implementation Engineer at Sutherland Healthcare Solutions

- Implemented EMR integrations across 15+ provider sites, enabling real-time HL7-based data exchange.
- Translated business and clinical requirements into HL7 interface designs and validation rules.
- Performed end-to-end testing, message validation, and data quality checks to ensure interoperability.
- Configured secure API and SFTP-based integrations using JSON and XML payloads.

Strategic EHR Interoperability & Vendor Alignment

HL7, Systematic Testing, Vendor Specification Analysis, Realtime Data Exchange, SQL

Implementation Engineer at Sutherland Healthcare Solutions

- Analyzed vendor specifications and designed HL7 interfaces aligned with interoperability workflows.
- Conducted validation of ADT, ORU, and DFT messages through structured QA processes.
- Coordinated with vendors and internal teams to resolve integration issues and ensure smooth delivery.
- Maintained SDLC documentation including requirement documents, test evidence, and change logs.

Healthcare Integration Pipeline (Mirth Connect & FHIR)

Mirth Connect 4.5.2, HL7 (ADT/ORU), JavaScript, XML, JSON, TCP/IP (LLP), File Polling (CRON), Custom Metadata, PDF Generation

Independent Technical Project

- Real-Time HL7 Pipeline: Built a functioning integration engine using Mirth Connect 4.5.2 to process ADT and ORU messages via TCP/LLP and File Polling.
- Data Transformation: Developed custom JavaScript logic for multi-format handling, enabling seamless translation between HL7, XML, and JSON payloads.
- FHIR Modernization: Mapped legacy clinical data to HL7 FHIR (R4) resource structures (Patient, Observation, Encounter) to simulate modern RESTful API-based data exchange.
- Validation & Monitoring: Implemented error-handling workflows to ensure data integrity across the integration pipeline.

EDUCATION

Kingston University London

London, UK

MSc Data Science – Merit

International Institute of Information Technology Bangalore

Bangalore, India

Postgraduate Diploma in Data Science – 3.29 / 4.0

NBN Sinhgad School of Engineering

Pune, India

BE Computer Science – 6.64 / 10.0